Literature Review Master Plan

AI Chatbot with Mood Detection — IT22638168

Pregnancy Support System (MaternaLink) — Project ID R26-IT-146

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Purpose of this document

This is a working research roadmap, not the literature review itself. It targets the current v2 architecture (Face + Text + Behaviour, behavioural signal under active review) and is designed to produce, by the end of the research phase, everything needed to write a defensible literature review and to resolve the open behavioural-signal decision on evidence rather than intuition.

A. Overall Literature Review Structure

Not every topic in a generic checklist earns a section here. The structure below is scoped to what this component actually needs to defend — each inclusion or exclusion is justified, not assumed.

#	Section	Include?	Reasoning
1	Maternal mental health & antenatal emotional wellbeing (global + Sri Lanka)	Yes — core	This is the problem justification. Sri Lankan-specific prevalence data is what Gap 1 stands on; already partly sourced in earlier revisions and needs completing, not starting from zero.
2	Existing pregnancy-support & maternal-health applications	Yes — core	Feeds directly into Section H (competitor matrix) and the Related Work chapter.
3	Conversational agents for mental health (chatbots)	Yes — narrowed	Narrow from generic 'AI/LLM healthcare chatbots' to perinatal/mental-health-specific agents (Woebot, Wysa, etc.) — broader healthcare-chatbot literature is not relevant to this component's claims.
4	Affective computing & Facial Emotion Recognition	Yes — core	Justifies Module 2's FER component and the MobileNetV2/TFLite technology choice; must include on-device/mobile FER and head-pose robustness as subsections, not separate chapters — scope discipline matters here.
5	Text sentiment analysis: English transformers + Sinhala/low-resource NLP	Yes — core, two subsections	Directly justifies DistilBERT and the custom Sinhala lexicon approach; Sinhala NLP needs its own subsection given how much this project's defensibility rests on it.
6	Multimodal emotion fusion	Yes — core	Justifies Module 3's fusion design and is where the 45/40/15 (or revised) weighting gets its methodological grounding.
7	Behavioural/passive signals for emotion inference	Yes — but flagged as under review	See Section G. This section's conclusion may end up justifying removal rather than inclusion — that is a legitimate literature review outcome, not a failure.
8	Adaptive / mood-conditioned conversational response generation	Yes — core	Justifies the actual mechanism (mood-conditioned system prompts) distinguishing this from a static chatbot.
9	Therapeutic music & digital interventions for antenatal anxiety	Yes — short	Grounds the music feature; does not need the same depth as the AI-technical sections since it supports one smaller sub-feature.
10	Privacy, ethics & on-device inference in health apps	Yes — short but necessary	Directly supports the consent-flow and on-device-processing design decisions already made; also pre-empts likely ethics-committee questions.
11	Existing systems comparative analysis	Yes — as its own chapter, not folded into #2	Needs to be a structured comparison table (Section H), not prose scattered through #2 — this is what makes Gap 1/2/3 visually obvious to an examiner.

Deliberately excluded or merged: a standalone 'mobile/edge AI' chapter (folded into FER as a subsection — not enough independent content to justify its own section at this scope); a standalone 'React Native / mobile development' chapter (this is engineering practice, not a research literature topic); a general 'LLM architecture' chapter (out of scope — the LLM itself is used, not studied).

B. Module-by-Module Research Plan

Module 1 — AI Chatbot

Field	Detail
Research topic	Conversational agents for mental health / perinatal support
Why needed	Justifies the choice of an LLM-based, mood-conditioned chatbot over a rule-based or scripted alternative
Key questions	What conversational-agent designs exist for perinatal/maternal mental health? What evaluation methods do they use? Do any support Sinhala or another South Asian language?
Papers	4–6
Ideal years	2019–2025 (chatbot/LLM field moves fast; anything pre-2018 is likely outdated for design comparison, though Woebot's original 2017 RCT is a legitimate exception as a foundational trial)
Source types	Peer-reviewed HCI/health-informatics journals, systematic reviews (e.g. perinatal digital-health reviews), primary app evaluation studies
Extract	Target population, language support, evaluation method, reported outcomes, whether adaptive/static
Influences	Objective 1 justification; Related Work comparison table

Module 2 — Mood Detection

B.1 Facial Emotion Recognition

Field	Detail
Research topic	FER architectures, transfer learning, and — critically — head-pose/viewing-angle robustness
Why needed	Justifies MobileNetV2 + TFLite; the head-pose subsection is not optional — it is the single most likely technical challenge in your viva given the phone-holding use case
Key questions	What accuracy do lightweight CNNs achieve on 3-class emotion output? How much does accuracy degrade at non-frontal head angles? What mitigations exist (confidence filtering, temporal smoothing, pose-aware training)?
Papers	5–7 (at least 2 must specifically address head-pose/viewing-angle effects — this is non-negotiable given the design decision already made in v2)
Ideal years	2018–2025, but foundational FER-2013/AffectNet papers (dataset origin papers) are correctly cited regardless of age
Source types	IEEE/ACM affective computing journals and conferences (FG, ACII), dataset papers, mobile-CV papers
Extract	Architecture, dataset, accuracy, class count, head-pose findings if present, real-world vs. lab conditions, inference speed on mobile hardware
Influences	§3.2.2 methodology, the confidence-threshold/EMA mitigation design, Appendix D risk R1

B.2 English Sentiment Analysis

Field	Detail
Research topic	Transformer-based sentiment classification, particularly distilled/lightweight models and negation handling
Why needed	Justifies DistilBERT over a lexicon approach for English — you already have the comparative reasoning; this section needs the citations behind it
Key questions	How do distilled transformers compare to full-size models on sentiment tasks? How well do general-purpose sentiment models transfer to health/wellbeing-context text outside their training domain?
Papers	3–4

Field	Detail
Ideal years	2019–2024 (DistilBERT itself is 2019; domain-transfer papers should be recent)
Source types	NLP conference papers (ACL, EMNLP), model/benchmark papers
Extract	Model size, benchmark accuracy, domain-transfer findings, negation-handling behaviour
Influences	§3.2.2 (English) technology-comparison paragraph, already partly written — this fills in the citation gap

B.3 Sinhala / Low-Resource Sentiment Analysis

Field	Detail
Research topic	Sinhala NLP resources, lexicon-based sentiment methods, low-resource language sentiment strategies generally
Why needed	This is one of your strongest defensible novelty threads — it needs to be the best-cited subsection in the whole review, not the thinnest
Key questions	What Sinhala sentiment resources exist and how were they built/validated? How does lexicon-based sentiment compare to model-based methods for genuinely low-resource languages? What domain-adaptation strategies exist for extending a general lexicon to a specific context?
Papers	5–7
Ideal years	2014–2024 (low-resource-language NLP is a smaller field; older foundational papers, e.g. Chen & Skiena 2014, remain correctly citable)
Source types	ACL Anthology, regional NLP conferences (ICTer, and similar Sri Lankan/South Asian venues), corpus/dataset papers
Extract	Language, method, dataset size, reported accuracy/coverage, whether data/lexicon is publicly available and under what licence
Influences	§3.2.3 methodology; directly informs how the domain-vocabulary extension is framed as original work

B.4 Behavioural Signals — see Section G (dedicated track, decision pending)

Module 3 — Adaptive Intelligence

Field	Detail
Research topic (fusion)	Multimodal emotion/affect fusion strategies — early vs. late fusion, modality weighting
Why needed	Justifies the fusion architecture itself and gives the weight values (45/40/15 or revised) a citable methodological basis instead of being asserted numbers
Key questions	How is modality weighting typically determined in affective-computing fusion systems — fixed, learned, or empirically tuned? Is visual-modality dominance in fusion an established finding or context-dependent?
Papers	4–5
Ideal years	2017–2024
Source types	Affective computing surveys/reviews (e.g. Poria et al.-style multimodal fusion reviews), fusion-methodology papers
Extract	Fusion strategy type, modality weighting method, reported performance vs. single-modality baselines
Influences	§3.2.6 methodology — the weighting justification paragraph

Field	Detail
Research topic (adaptive response)	Mood-conditioned or affect-aware response generation in conversational systems

Field	Detail
Why needed	Justifies the actual mechanism that makes this project's contribution distinct from a static chatbot with a bolted-on mood display
Key questions	Does conditioning response tone on detected affect improve engagement or perceived support in existing studies? How is this typically implemented (prompt injection vs. fine-tuning vs. rule-based)?
Papers	3–4
Ideal years	2020–2025 (LLM-prompt-based approaches are recent by necessity)
Source types	HCI/conversational-AI conference papers, affective dialogue system papers
Extract	Conditioning mechanism, evaluation method, reported effect on user experience
Influences	§3.2.6/Objective 3 justification

Module 4 — Data Management

No dedicated literature section is needed here — offline-first mobile storage and JSON export are engineering practice, not an open research question. The relevant citation need is a short one covering offline-first design in low-connectivity health contexts (1–2 papers), folded into the Sri Lankan background section rather than given its own chapter.

C. Paper Collection Target

This is an undergraduate proposal-level literature review supporting one component of a group project — not a standalone systematic review or a PhD-level survey. Volume for its own sake actively hurts you: a bloated reference list with weak per-paper engagement reads worse than a tight, well-extracted one.

Level	Count	Interpretation
Minimum useful	18 papers	Below this, several sections (Sinhala NLP, head-pose FER, fusion weighting) will not have enough support to defend the specific claims already made in the v2 methodology.
Recommended	26–30 papers	Enough depth per topic to write a genuine synthesis (not just a list) while remaining achievable in the 2-week literature review WBS task alongside the rest of your workload.
Maximum before diminishing returns	~35–40 papers	Beyond this, additional papers mostly duplicate findings already covered; time is better spent on deeper extraction of existing papers or on the extra existing-systems analysis in Section H.

Distribution across topics (recommended-level target: ~28)

Topic	Target papers
Maternal mental health / Sri Lanka context	4–5
Existing pregnancy-support & mental-health apps (feeds Section H too)	2–3 (plus the systems reviewed directly in Section H, not double-counted)
Conversational agents for mental health	4–6
Facial Emotion Recognition (incl. ≥2 on head-pose specifically)	5–7
English sentiment / transformers	3–4
Sinhala / low-resource NLP	5–7
Multimodal fusion	4–5
Adaptive/affect-conditioned response generation	3–4
Therapeutic music & digital interventions	2–3
Privacy/ethics/on-device inference	2–3
Behavioural signal track (Section G)	4–6, tracked separately — see below

Note: topic targets overlap slightly by design (a paper can legitimately count toward two subsections, e.g. a Sri Lankan telehealth-privacy paper). Do not force artificial separation just to hit per-topic quotas.

D. Source Quality Strategy

Tier	What counts	Examples relevant to this project
Tier 1	Systematic reviews, meta-analyses, papers in major peer-reviewed journals or top-tier conferences	IEEE Transactions on Affective Computing, ACL/EMNLP main proceedings, BMC Psychiatry, Ceylon Medical Journal, JMIR Mental Health
Tier 2	Strong single-domain studies, technical/method papers, validated public datasets and model cards	AffectNet/FER-2013 dataset papers, MobileNetV2/DistilBERT original papers, Hugging Face model cards with reported benchmarks, regional conferences (ICTer)
Tier 3	Supporting sources — theses, preprints, industry reports used only for context, not as primary evidence for a technical claim	TRCSL statistical reports, Department of Census and Statistics reports, well-documented GitHub repos accompanying a paper

Avoid or use only with extra caution

- Any source that cannot be independently verified to exist and say what it's cited as saying — given this exact project's earlier history with fabricated citations, this is not a generic warning, it is a mandatory check on every single source before it enters the extraction sheet.
- Blog posts, marketing pages, and un-refereed preprints as primary evidence for a technical accuracy claim — acceptable only as pointers toward a real paper, never as the citation itself.
- Predatory or non-indexed journals — if a journal doesn't appear in a recognizable index (Scopus, Web of Science, IEEE Xplore, ACL Anthology, PubMed) and isn't a known Sri Lankan academic venue, verify its legitimacy before using it.
- Google Scholar search-result links used as the citation itself, rather than the actual paper/DOI — a repeat of the exact problem found and fixed earlier in this project.

E. What to Extract From Every Paper

Use this as the column headers of the Paper Extraction Sheet (Section I). Every paper gets one row — no exceptions, including papers you're fairly sure you won't end up citing, since a fast 'not relevant, discarded' entry is cheaper than re-finding a paper later.

Field	What goes here
Citation (full)	Author, year, title, venue — enough to build a real IEEE-format reference, not a search link
DOI / stable link	Direct link to the actual paper, verified to load
Research problem	One sentence: what question the paper is answering
Dataset	Name, size, source, licence if stated
Population / context	Who the subjects were, what country/language/setting
Method	Study design or technical method in brief
Model/algorithm	Specific architecture or technique, if applicable
Input modality	Face / text / behaviour / audio / other
Output	What the system classifies or predicts
Evaluation metrics	Accuracy, F1, precision/recall, prevalence %, etc.
Results	The actual numbers, not just 'performed well'
Limitations (stated by authors)	What the paper itself admits it doesn't solve
Relevance to IT22638168	Which specific section/module/decision this supports
Gap identified	What this paper does NOT cover that your project addresses
Technology/technique to reuse or reference	Concrete takeaway — a number, a method name, a design pattern

Recommended format

Build this as a spreadsheet (one row per paper), not a Word document — you'll be sorting/filtering by topic and by relevance constantly during writing. Word is the right format only for the final literature review chapter itself.

F. Research Gap Strategy

The funnel, applied consistently to every candidate gap

For each gap area below: state what the existing research actually shows → identify the specific limitation the existing research itself admits or that becomes visible by comparing several papers → state the resulting unresolved problem → state the gap in one sentence → connect it to what IT22638168 specifically does about it. Do not skip from 'existing research' straight to 'gap' — the limitation step is what makes a gap claim defensible rather than asserted.

Candidate gaps to investigate — not assumed to exist

Candidate gap	What to actually check in the literature before claiming this
Sri Lankan pregnancy emotional-support gap	Do the Sri Lankan antenatal depression/anxiety prevalence studies actually report an absence of routine screening? (Already partly confirmed in earlier revision work — needs the remaining citations completed, not re-litigated from scratch.)
Sinhala/English bilingual support gap	Search specifically for any Sinhala-language mental-health or wellbeing chatbot, published or deployed, before asserting none exists — a negative claim needs an active, documented search, not just an absence of results you happened to find.
Passive multimodal mood detection gap (South Asia)	StandStrong (Nepal) is already known to exist and must be checked again for any newer related work — confirm it remains phone-sensor-based rather than chatbot-embedded facial+text fusion before repeating the narrowed Gap 2 claim.
Face + text fusion specifically (as opposed to any multimodal fusion)	Multimodal fusion combining vision and NLP is well studied generally — check whether the gap claim needs to be about the *application domain* (perinatal chatbot) rather than the fusion technique itself, which is not novel in isolation.
Adaptive/mood-conditioned response in a pregnancy-specific chatbot	Check the perinatal-chatbot review literature (e.g. the Haga et al. 2021-style review, updated with anything since) specifically for whether any reviewed system adapts tone/response based on detected mood, not just whether any system exists.
On-device FER for privacy in a health app	This is more of a design-rationale citation than a gap claim — check whether on-device inference is established practice in health-app privacy literature, to support the choice rather than to claim novelty in doing it.

Important discipline

Every gap claim in this project must survive the question 'what did you search to confirm nobody has done this?' A gap claim with no documented search behind it is exactly the kind of thing an examiner who knows the space will puncture in thirty seconds.

G. Behavioural Signal Decision — Dedicated Literature Track

This track exists specifically to resolve the open architecture question with evidence, not to justify a decision already made. Run this track in parallel with the rest of the review, but resolve it before finalizing §3.2.6 methodology, since the fusion equation depends on its outcome.

Sub-topic	Specific question to answer	Papers
Typing speed & emotion	Is typing speed shown to predict mood at the individual/message level, or only as a population-level statistical association?	1–2
Response delay & emotion	Same individual-vs-population distinction, specifically for inter-message response latency	
Keystroke/behavioural affect detection generally	What's the best-reported accuracy for behaviour-only affect classification, and under what controlled conditions?	1–2
Individual-level vs. population-level inference	This is the single most important question — does any paper validate behavioural signals as a real-time, per-instance classifier (with a reported accuracy/F1), or is all the evidence correlational at the group level?	covered by the above, cross-checked explicitly
Device/keyboard confounds	Has any study controlled for device type, keyboard/IME, or connection speed when measuring typing-based affect signals?	1
Sinhala vs. English typing differences	Is there any published baseline for Sinhala Unicode typing speed vs. English, independent of mood, that would explain away a false signal?	0–1 (may not exist — if it doesn't, that absence itself is a finding worth stating)

Decision rule — apply after collecting this track

1. If at least two Tier 1/2 sources demonstrate individual-level, per-instance validated behavioural affect classification with a reported accuracy metric → KEEP in fusion, with that metric used to justify a defensible weight.
2. If the evidence found is exclusively population-level/correlational, with no individual-level validated classifier → REMOVE from the fused mood score; retain as logged, unfused telemetry only, citing the population-level literature as rationale for why it's collected but not trusted as a per-message input.
3. If evidence is genuinely mixed or thin either way → default to telemetry-only (option 2), since introducing an unvalidated signal into a mechanism that shapes responses to a vulnerable population carries more downside than the upside of keeping a weak signal in fusion.

Given what's already known from earlier work on this project (Giannakakis et al. 2022 is population-level, not individual-level), the working expectation going in is that this track resolves toward option 2 or 3 — but the literature search should be run properly rather than treated as confirmatory.

H. Existing System / Competitor Review

Target: 9–12 systems total, enough to populate a genuinely comparative matrix without padding it with near-duplicates.

Category	Systems to include	Count
Perinatal/pregnancy-specific chatbots	Any dedicated pregnancy mental-health chatbot found (may be very few or none — that absence is itself part of Gap 1/2)	1–2
General mental-health chatbots	Woebot, Wysa, and one more if a strong third candidate exists	2–3
General pregnancy-tracking apps	Glow, BabyCenter, or regional equivalents	2
Sri Lankan telehealth platforms	oDoc, ChannellingLK, or similar	1–2
Passive-sensing maternal-health systems	StandStrong (Nepal) and any comparable South Asian/low-resource system found	1–2
Sinhala/low-resource-language digital health or NLP systems	Any Sinhala-language health or wellbeing app, even outside pregnancy specifically	1–2

Comparison matrix columns

Language support · Pregnancy specialization · Mood/emotion detection (yes/no, modality) · Modalities used · Adaptive response (yes/no) · Privacy model (on-device vs. cloud) · Offline capability · Therapeutic content · Clinical/doctor integration.

This table is what makes Gap 1–3 visible without you having to argue for them in prose — a mostly-empty column under 'Sinhala' and 'Adaptive response' across every row does the argument's work for you.

I. Literature Review Document Plan

Six documents — not twenty. Each has a distinct job; merging any two would blur working material with final dissertation material, which makes both harder to maintain.

Document	Purpose	Contains	Created	Type
1. Literature Search & Scope Plan	This document — the roadmap itself, updated if scope changes	Sections A–H content	Now, before searching begins	Working
2. Paper Extraction Sheet (spreadsheet)	One row per paper, using the Section E template	All fields from Section E, for every paper screened (including rejects, marked as such)	Continuously, during search	Working
3. Existing Systems Comparison Matrix	The Section H table, fully populated	9–12 systems × 9 comparison columns	After Section H's systems are identified	Working → becomes a table directly reused in the Related Work chapter
4. Research Gap Matrix	Applies the Section F funnel per gap, with citations	One row per candidate gap: existing research / limitation / unresolved problem / gap statement / how IT22638168 addresses it	After extraction sheet has enough coverage per topic	Working → source material for the Research Gap section of the final chapter
5. Behavioural Signal Decision Memo	Short, standalone record of the Section G track and its outcome	The 6 sub-questions, papers found, decision rule applied, final KEEP/REMOVE/TELEMETRY-ONLY decision with reasoning	After Section G track is complete, before finalizing §3.2.6 methodology	Working → decision gets cited in the final proposal/methodology, memo itself stays as your own audit trail
6. Final Literature Review (dissertation chapter)	The actual written chapter	Synthesized prose following the Section A structure, drawing on documents 2–5	Last — after 1–5 are substantially complete	Final dissertation material

J. Workflow

4. Confirm/refine research questions per module — use Section B's "key questions" as the starting list; adjust only if a question turns out to be unanswerable or irrelevant once searching starts.
5. Build search keyword sets per topic (e.g. FER: "facial expression recognition head pose accuracy", "MobileNetV2 emotion classification"; Sinhala NLP: "Sinhala sentiment lexicon", "low-resource language sentiment analysis") — reuse the exact search terms that already worked earlier in this project where applicable.
6. Search and collect candidate papers against the Section C targets, logging every candidate (including rejects) into the Paper Extraction Sheet.
7. Screen each candidate against Section D's tier definitions; verify every source actually resolves to a real, checkable document before extraction — this step is not optional given this project's history.
8. Extract full detail (Section E template) for every paper that passes screening.
9. Run the Behavioural Signal track (Section G) in parallel with steps 3–5; produce the Decision Memo (Document 5) as soon as that track has enough coverage, since it gates the fusion methodology.
10. Build the Existing Systems Comparison Matrix (Document 3) once enough systems are identified — this can run concurrently with steps 3–5.
11. Once each topic has adequate coverage, apply the Section F funnel per candidate gap and build the Research Gap Matrix (Document 4).
12. Cross-check: does every gap claim in Document 4 map to a specific decision already made in the v2 architecture/methodology? If a gap doesn't connect to anything the system actually does, either the gap is misstated or the system is missing something — resolve the mismatch before writing.
13. Finalize the behavioural-signal architecture decision (fusion equation in §3.2.6) using the Decision Memo's outcome.
14. Draft the final Literature Review chapter (Document 6) following the Section A structure, drawing directly from Documents 2–5 rather than re-deriving arguments from memory.
15. Final verification pass: every citation in the chapter must trace back to a real row in the Extraction Sheet with a working DOI/link — no citation should exist in the final chapter that isn't backed by an extraction-sheet row.

K. Final Deliverable — What You Should Have When This Phase Is Done

- 26–30 (or a defensible number between 18–40) fully extracted, verified papers in the Extraction Sheet, distributed roughly per the Section C table.
- A completed Research Gap Matrix with each gap claim traceable through the existing research → limitation → gap funnel, not asserted.
- A completed Existing Systems Comparison Matrix (9–12 systems) ready to drop into the Related Work chapter.
- A one-paragraph, citation-backed gap statement for each of Gap 1, 2, and 3 (or however many survive the F funnel — it's fine if one candidate gap doesn't hold up and gets dropped).
- A technology-justification paragraph for each major choice (MobileNetV2/TFLite, DistilBERT, the Sinhala lexicon approach, the fusion strategy) grounded in the extracted literature rather than in this conversation's reasoning alone.
- A completed Behavioural Signal Decision Memo with a clear KEEP / REMOVE / TELEMETRY-ONLY outcome and the evidence behind it — this then updates §3.2.6 and the fusion weights in the proposal.
- A final architecture-justification statement: one paragraph per module explaining, with citations, why each component was chosen — this is what actually gets defended in the viva, and it should read as evidence-led rather than as a list of technologies.
- A complete Literature Review chapter following the Section A structure, ready to insert into the full proposal/dissertation at the appropriate section numbers.

One last check before this phase counts as done

Read the finished chapter once specifically hunting for any claim, number, or citation that isn't backed by a row in the Extraction Sheet. Given this project's documented history of fabricated references earlier in its development, this single pass is the highest-value five minutes in the entire literature review phase.